

Applying spatio-temporal analysis for data mining on shooting data

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1. Introduction

The Fogo Cruzado Institute employs technology to produce and disseminate open and collaborative data on armed violence, thereby strengthening democracy through social transformation, life preservation, and supporting the improvement of public security policies. Through a mobile application, and active research in social and news media, Fogo Cruzado receives and organises information on shootings, verified in real-time by a specialised local team utilising an innovative methodology in the metropolitan regions of Rio de Janeiro, Recife, Salvador, and Belém in Brazil. This information is accessible in the first open database on armed violence in Latin America, available through the Institute's API.

The Fogo Cruzado database hosts more than 56,380 shooting occurrences crowdsourced from the public. Daily, it is estimated that more than 18 new records are created by collaborators. These records include various details such as the presence of victims and whether the incident is related to police action, composing 40 indicators in total. Given that shooting data is mapped as points with longitude and latitude coordinates, also known as point processes (Bivand, Pebesma, and Rubio, 2008), traditional spatial analysis methods can be applied to examine their spatial distribution and structure. First-order analysis, such as kernel density estimation (KDE), assesses the distribution of events across the study area, providing insight into how shooting occurrences are distributed throughout the territory. Second-order analysis, like Ripley's K-function or L-function (Ripley, 1976; Waller and Gotway, 2004), investigates the spatial structure of events, revealing whether they occur randomly, regularly, or in clusters.

Although first- and second-order analyses provide valuable insights into the spatial distribution and structure of events and have been used successfully as a data mining tool (Silva et al.), these approaches consider only the spatial dimension and overlook the temporal aspect, which can lead to misinterpretation of the data by missing critical patterns and trends that are essential to understanding the dynamics of shooting incidents.

Analysing shootings using spatiotemporal methods has the potential to enhance our understanding by integrating both temporal and spatial components, rather than examining them in isolation. This approach, when applied as a data mining tool on datasets like Fogo Cruzado's, can reveal recurrent territorial conflicts where events recur both spatially and temporally, providing valuable insights into the nature of gun assaults. These insights are crucial for generating policy-relevant information that informs policy and guides the allocation of limited resources within the criminal justice system and crime prevention efforts, ensuring a more effective and targeted response to gun violence.

2. Methodology

The Knox local spatio-temporal correlation approach is a powerful tool that incorporates both spatial and temporal attributes. Originally developed in epidemiology to study disease outbreaks, the Knox statistic identifies clustering of events in space and time using distance thresholds (Δ) for spatial neighbours and time thresholds (τ) for temporal neighbours. Since τ can be any temporal measure, we used 5 days as the cutoff temporal distance from shootings to be identified as temporally correlated, while we used 300 metres as the Δ threshold.

From both thresholds, Knox statistics identifies whether there is a departure from the null hypothesis (global approach) and which cases could be considered spatiotemporally correlated with a departure from the null hypothesis. The null hypothesis of the Knox test is that there is no spatiotemporal interaction between the observed events. If more events than expected are found within the defined spatial and temporal bounds (i.e., more neighbours within the spatiotemporal window than expected), this provides evidence to reject the null hypothesis, indicating spatiotemporal clustering in the observed events. To be considered spatiotemporally correlated, a shot must occur within five days of each other and within 300 metres of each other.

To test the use of Knox spatio-temporal correlation as a data mining tool, we applied it to crime data from the 13 cities of the metropolitan region of Salvador, Bahia, Brazil. By measuring time in days since data collection began, we utilised the Knox statistic to identify space-time neighbours and assess their significance through both analytical and simulation-based inference. Observations were marked as hotspots if the number of space-time neighbours exceeded the expected number, with each observation assigned a p-value to determine its significance.

3. Results

To evaluate the effectiveness of Knox spatio-temporal correlation as a data mining tool, we utilised the Python *pointpats* implementation (Kang et al., 2023) and applied it to Fogo Cruzado's data from the 13 cities in the metropolitan region of Salvador. Shooting incidents range from the beginning of Fogo Cruzado's mapping of the region (2022-07-01) to 2024-06-30, with a total of 3473 incidents. Since the mapping process uses the 3857 WGS84 - Pseudo Mercator, all occurrences were projected to 31984 SIRGAS 2000 UTM Zone 24 South so that all distance-related measurements could be made in metres with low errors.

City name	Total shootings
Salvador	2566
Camaçari	354
Lauro De Freitas	143
Simões Filho	93
Mata De Sao Joao	77
Dias D'ávil	76
Vera Cruz	45
Candeias	43
Madre De Deus	27
São Sebastião Do Passé	16
Itaparica	12
São Francisco Do Conde	12
Pojuca	9

Table 1: Cities in the metropolitan region of Salvador with their total shootings recorded from July 1, 2022, to June 30, 2024..

In the global Knox spatio-temporal correlation test, only three cities had a p-value less than 0.05: Salvador, Camaçari and Dias D'ávil (table 2). This significant deviation from the null hypothesis suggests that observed shootings in these cities tended to cluster in both space and time.

City	Shootings	p-value
Salvador	2566	0.01
Camacari	354	0.01
Dias D'avila	76	0.01

Table 2: Cities in the metropolitan region of Salvador with a deviation from the null hypothesis, indicating a systematic spatiotemporal correlation in the Knox Global Test.

The global Knox spatio-temporal test is an effective starting point for identifying cities with systemic and recurring clusters of shootings. However, the local Knox statistic provides a more detailed analysis by identifying spatio-temporal clusters at the individual event level. This approach allows us to determine for each observed shooting event whether it is a lead point, a lag point, or a coincident point. Lead points are events that spatiotemporally influence other dependent events. Lag points are those that are spatiotemporally dependent on preceding lead points. Coincident points are events that occur simultaneously with others without a clear lead or lag relationship.

Using the Knox Local statistic as a data mining tool with the full dataset, 187 shooting events were identified as spatiotemporally clustered with a significant deviation from the null hypothesis, of which 40 were identified as "lead", 100 as "lag", and 45 as "coincident". This suggests that these 185 observations are likely related to conflict rather than sporadic shooting, highlighting a potential deeper problem.

Salvador had 30 leading shooting observations with 92 dependent events, followed by the city of Camaçari with 8 leading observations and 6 lagging observations. Dias D'Ávila had 1 leading observation with 3 correlated events (table 3, figure 1). Additionally, São Sebastião do Passé had one leading and one dependent shooting observation, although it did not deviate from the null hypothesis in the Knox Global analysis.

City name	Observation status	n
Salvador	lead	30
	lag	92
	coincident	43
Camaçari	lead	8
	lag	6
	coincident	2
Dias D'ávil	lead	1
	lag	3
São Sebastião Do Passé	lead	1
	lag	1

Table 3: Knox local statistics showing the number of lead, lag, and coincident shooting observations for each city.

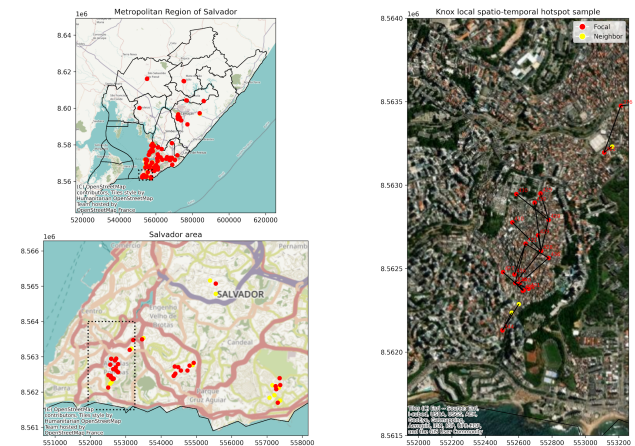


Figure 1: Maps of Salvador Metropolitan, Salvador and a sample of the Knox local spatio-temporal hotspot.

The Knox local spatio-temporal analysis successfully highlighted areas and periods with significant space-time clustering of shooting events, differentiating recurring incidents from sporadic occurrences. This approach provides a more comprehensive understanding of territorial conflicts and can generate alerts for recurring violence, offering valuable insights for public security policies.

All data and information required for reproducibility are available in [this repository](#). The ability to reproduce our findings highlights the robustness of the methodology, demonstrating its effectiveness as a valuable tool for data mining in the study of armed violence.

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